A Study in Voice Biometrics

The Effects of Denoising on Speaker Recognition

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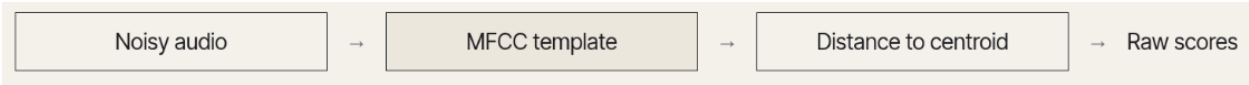
1. *Motivation & Background*

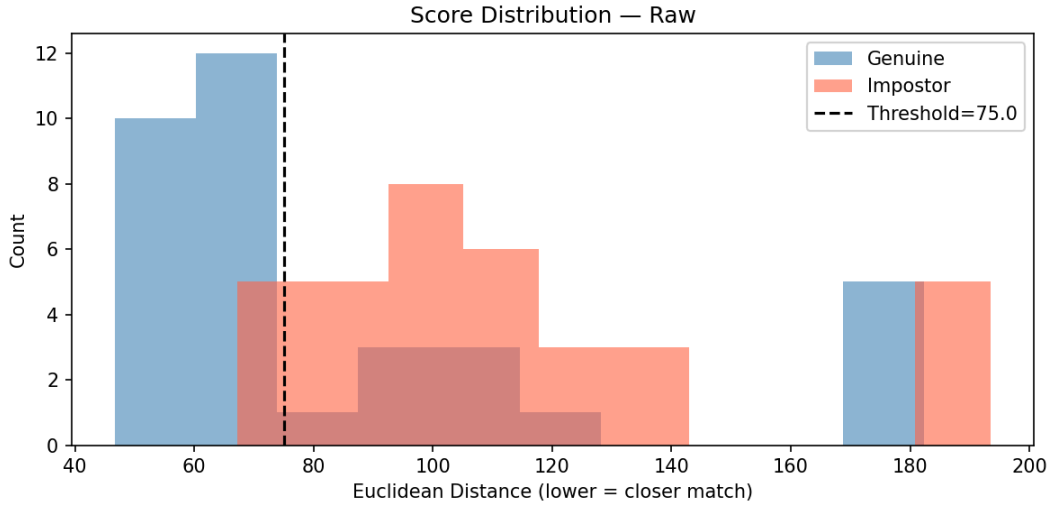
Voice is a widely used biometric, but unlike fingerprints or iris recognition, it is often collected in uncontrolled environments. This can introduce confounding variables that negatively affect speaker recognition systems and algorithms. As you add noise to a sample it trends the vector representing each speaker's voice towards a high-entropy centroid representing the noise environment. As this effect is applied jointly to both the true samples and impostors it shrinks the gap in distances between genuine and impostor distributions. Thus in an effort to better differentiate the voices of speakers out in the wild we are looking into the effects of denoising on speaker recognition.

Speaker recognition systems work with the frequency content of voice signals in recorded clips. And then those analyses are used by a wide variety of systems to perform distance calculations and distinguish speakers. These methods can include but are not limited to pulling out spectral and or waveform characteristics and embedding them in a vector space, training neural networks to discriminate sample waveforms, as well as any number of other bespoke separation methods. What all of these share is that they are working off of a polluted sample. All environments and collection methods stand to introduce noise into the sample, and in myriad ways. This results in a space that is overcome by confounding variables which inhibit speaker separation.

MFCCs (Mel Frequency Cepstral Coefficients), take the waveform and extract the power spectrum on frequencies commonly used in human speech. These can then be used for voice matching using a pipeline like below.



These raw scores can then be used to establish the fitness of a given function for biometric identification. Arranged in a histogram we can see visually how hard it is to distinguish voices from noisy waveforms.



Fitness functions like FAR (False Accept Rate), FRR (False Reject Rate), and TAR (True Accept Rate) can then be mapped out to show how effective the given system is.

2. *Methods*

We implemented two different pipelines: a raw and a denoised. The processes follows as:

Track A (Raw): Noisy audio → MFCC → Distance → Classification

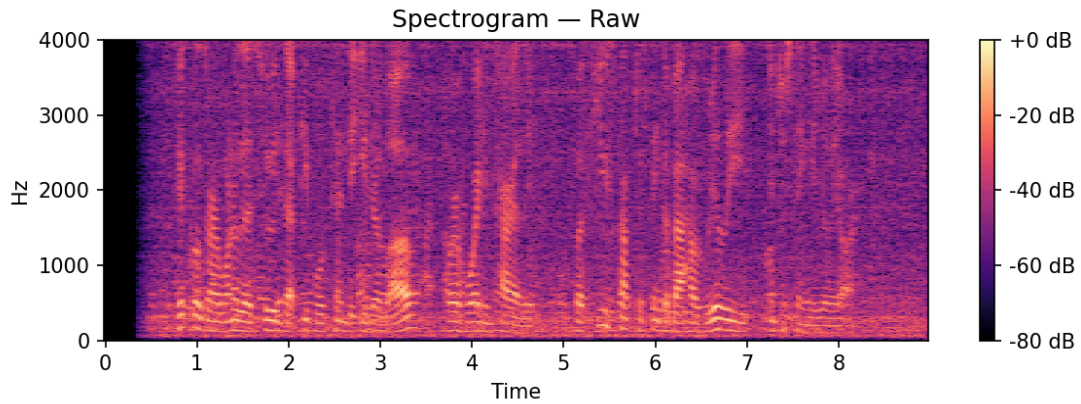
Track B (Denoised): Noisy audio → Denoised → MFCC → Distance → Classification

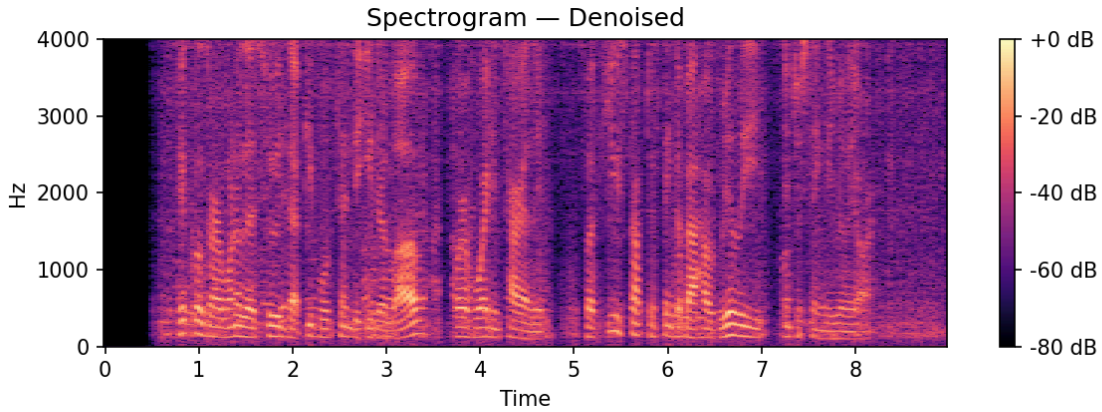
Our data set consisted of 80 samples and two speakers, both speakers enrolled with raw data composed of 5 audio clips each, noisy samples were then collected all across University of North Carolina at Charlotte's campus. We took a broad sample of noise environments from library common areas, construction zones, roadsides, HVAC systems and other indoor and outdoor locations, totaling 70 noise samples. Each sample was 6-9 seconds long featuring a speaker reading one of 5 sentences in the environment. Each environment was used to record 10 samples, 5 from each speaker.

Once the data was collected we used librosa to compute 13 MFCCs per frame across each clip, then summarized each coefficient across all frames using its mean, standard deviation, and mean delta, producing a 39-dimensional feature vector per clip. We use those vectors for a Euclidean distance measure. When enrolling a new

speaker we took the centroid of 5 of these vectors collected from quiet-ish samples to help remove confounding variables in the enrollments.

With this data and speaker recognition system in place we then worked with Jahangir Khondkar to utilize his denoising system for our test. As is shown in the spectrograms below, what was originally a very noisy sample with hard to isolate voice features has been significantly cleaned up and now shows clear voice lines.

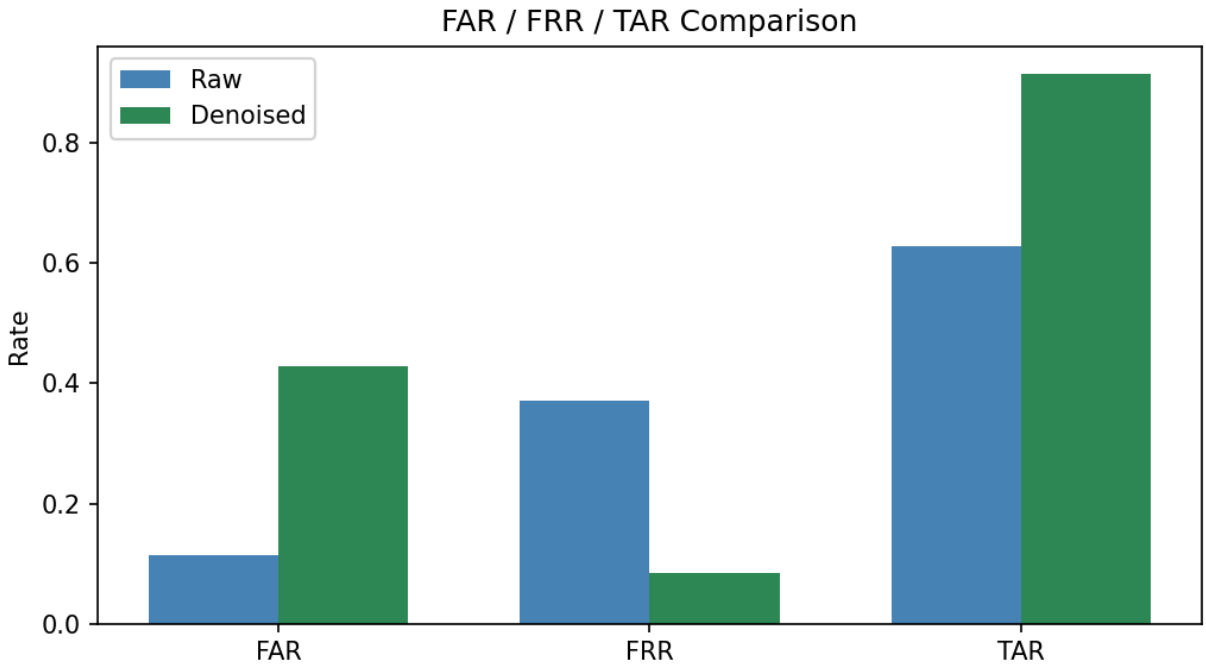




With this complete it was a simple matter of running our entire dataset through the system to get some statistical outputs on our FAR, FRR, and TAR.

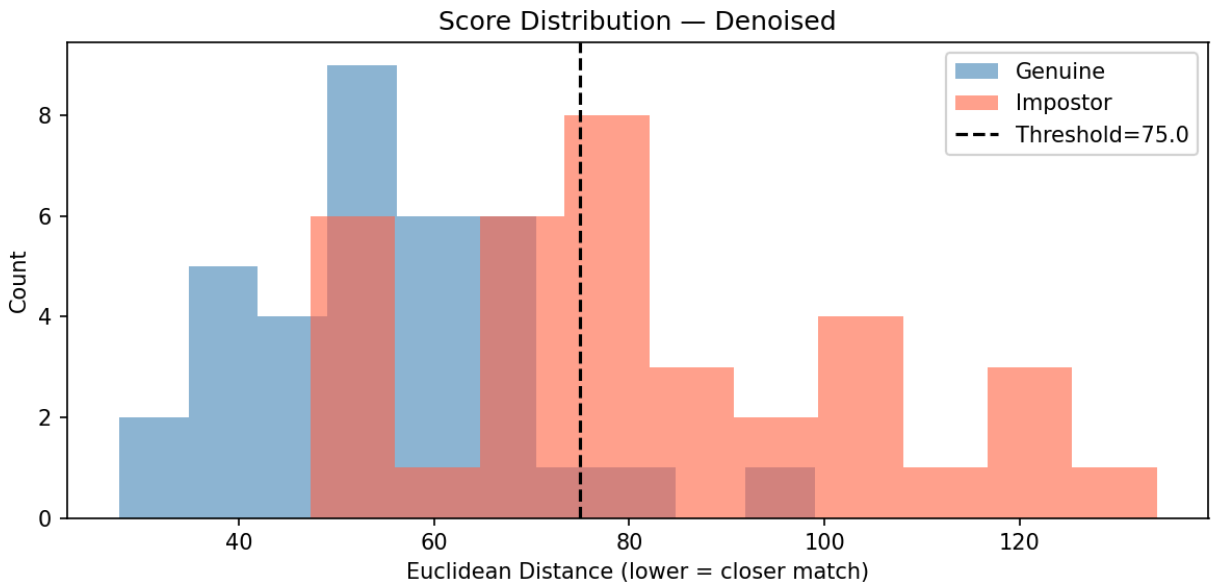
3. Results

Denoising had a clear impact on the performance of our speaker recognition algorithm. The most significant improvement was observed in our TAR which increased from 0.63 → 0.92. Indicating our system became much better at correctly identifying speakers post denoising. Similarly, our FRR dropped significantly from 0.37 → 0.08, meaning fewer legitimate users were incorrectly rejected. That being said it wasn't all sunshine and rainbows, our FAR had increased from 0.11 → 0.43.



This can also be observed in the way our histogram now shows better differentiated distributions for our genuine and impostor sample sets. The threshold was left set at 75 as that

was an optimal value for the initial set, though lowering that threshold in the denoised results would likely yield improved metrics.



4. Conclusion / Discussion

Our findings, while showing clear movement in the score distributions, are limited by the small scale of the dataset — only two speakers with 35 samples each. Broader conclusions would require additional speakers, discrimination techniques, and denoising systems.

In conclusion, denoising improved speaker recognition performance by moving the genuine and impostor samples into a far more normal distribution than the noisy samples provided, and by increasing the separation between those clusters. This can be seen in the improvements to the TAR and FRR. However, it did not fully separate the clusters as is evidenced in the coinciding increase in the false accept rate when using a fixed threshold. This

could be improved in future systems by better tuning the threshold, though perfect separation would likely require a more discriminating algorithm or more aggressive denoising.

5. Appendix

Code: https://github.com/KingKutza/DR-Voice-Detect — includes the MFCC extraction pipeline, enrollment routine, and scoring scripts.